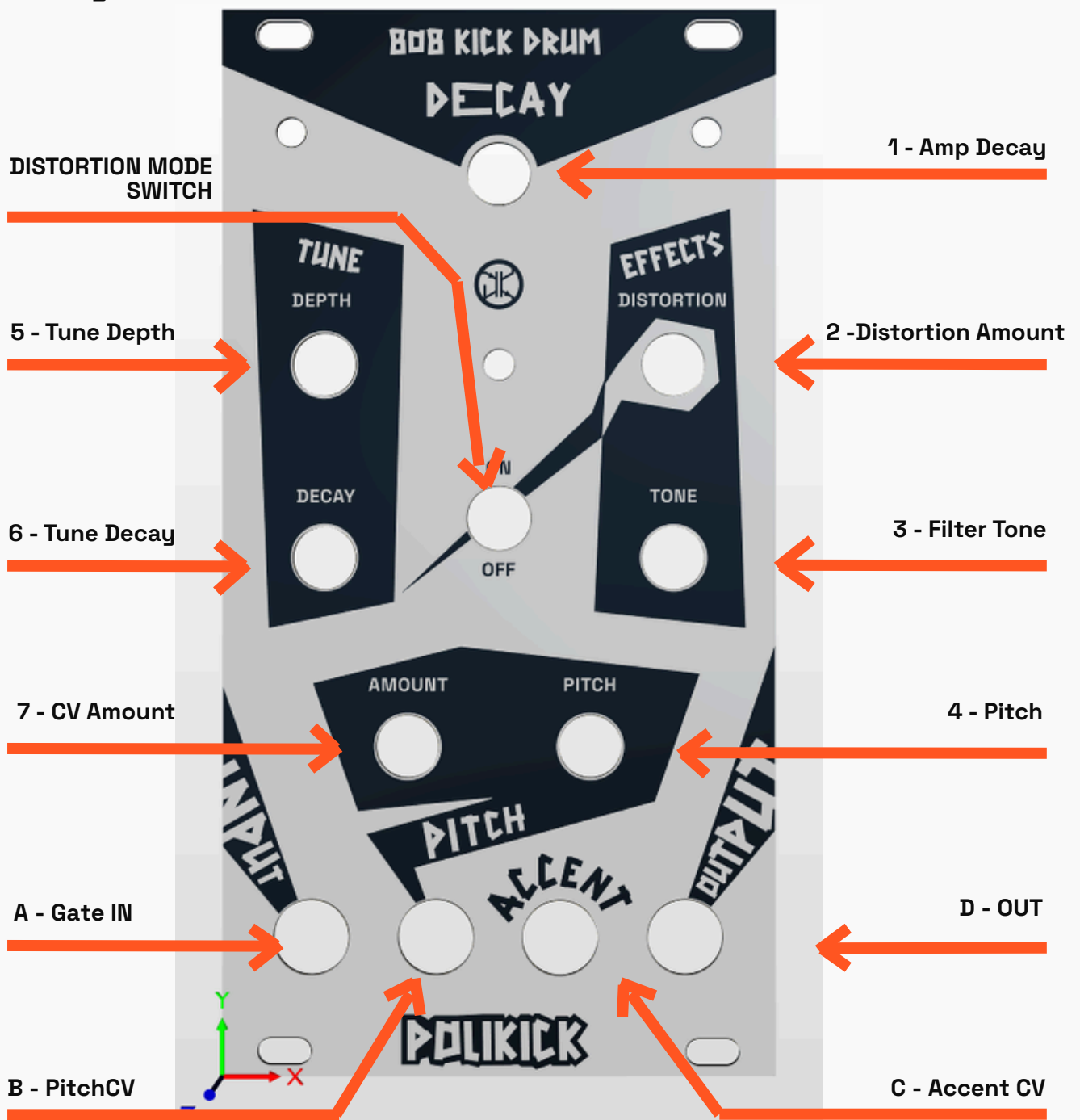


DATASHEET POLIKICK

Analog Eurorack KickDrum developed by PoliTeK
Student Team - Polytechnic University of Turin

OVERVIEW

The **PoliKick** is an analog kick drum designed to be integrated into a Eurorack modular system. Inspired by the iconic Roland TR-808, the module adapts the historic design to the needs and possibilities of the modular format. The circuit takes inspiration from the Erica Synths EDU DIY kick drum, adding a selectable distortion stage and an extended note range.



POTENTIOMETERS DESCRIPTIONS

#	NAME	DESCRIPTION
1	Amp Decay	Controls the length of the sound. Turn it down for a short, tight click, or turn it up for a long, booming 808-style tail
2	Distortion Amount	Controls the amount of saturation and drive when the distortion stage is engaged.
3	Filter Tone	Active low pass filter cutoff frequency, shapes the timbre of the kick drum.
4	Pitch	Adjusts the fundamental frequency of the kick drum. It ranges from deep, sub-bass thuds to higher, punchy tom-like tones.
5	Tune Depth	Sets the depth of the initial pitch envelope, controlling how much the pitch drops at the start of the hit for extra punch.
6	Tune Decay	Controls the speed of the pitch envelope, adjusting how fast the initial pitch drop occurs.
7	CV Amount	Attenuates the level of the incoming Control Voltage (CV) to control the external modulation voltage effect.

JACK IN/OUT DESCRIPTIONS

#	NAME	TYPE	DESCRIPTION
A	Gate	Input	Connect a gate here from a sequencer or clock source to fire the kick drum. Minimum Voltage to trigger is ~2.5V
B	Pitch	Input	Control Voltage input for pitch modulation. Allows you to change the fundamental frequency externally. Not V/oct
C	Accent	Input	Control Voltage to dynamically modulate the volume of the hit, making specific steps stand out in a sequence.
D	Output	Output	The main audio output. Delivers the final analog kick drum signal to your mixer, VCA, or output module.

ELECTRICAL SPECIFICATIONS

PARAMETER	TYPICAL VALUE
Supply voltage	+ - 12V DC
Current consumption	< 100 mA
Input impedance	100 k Ω
Output impedance	1 k Ω
Max Input Voltage	12 V

CONNECTORS & LAYOUT

- Input/Output Jack: 3.5 mm
- Power Supply: 12V DC 5 pins flat cable

RECOMMENDATIONS

- Do not exceed 12V power supply
- Use stabilized power supplies to avoid noise.

CONTACTS & CREDITS

For info or support:
<https://github.com/PoliTeK/PoliKick>